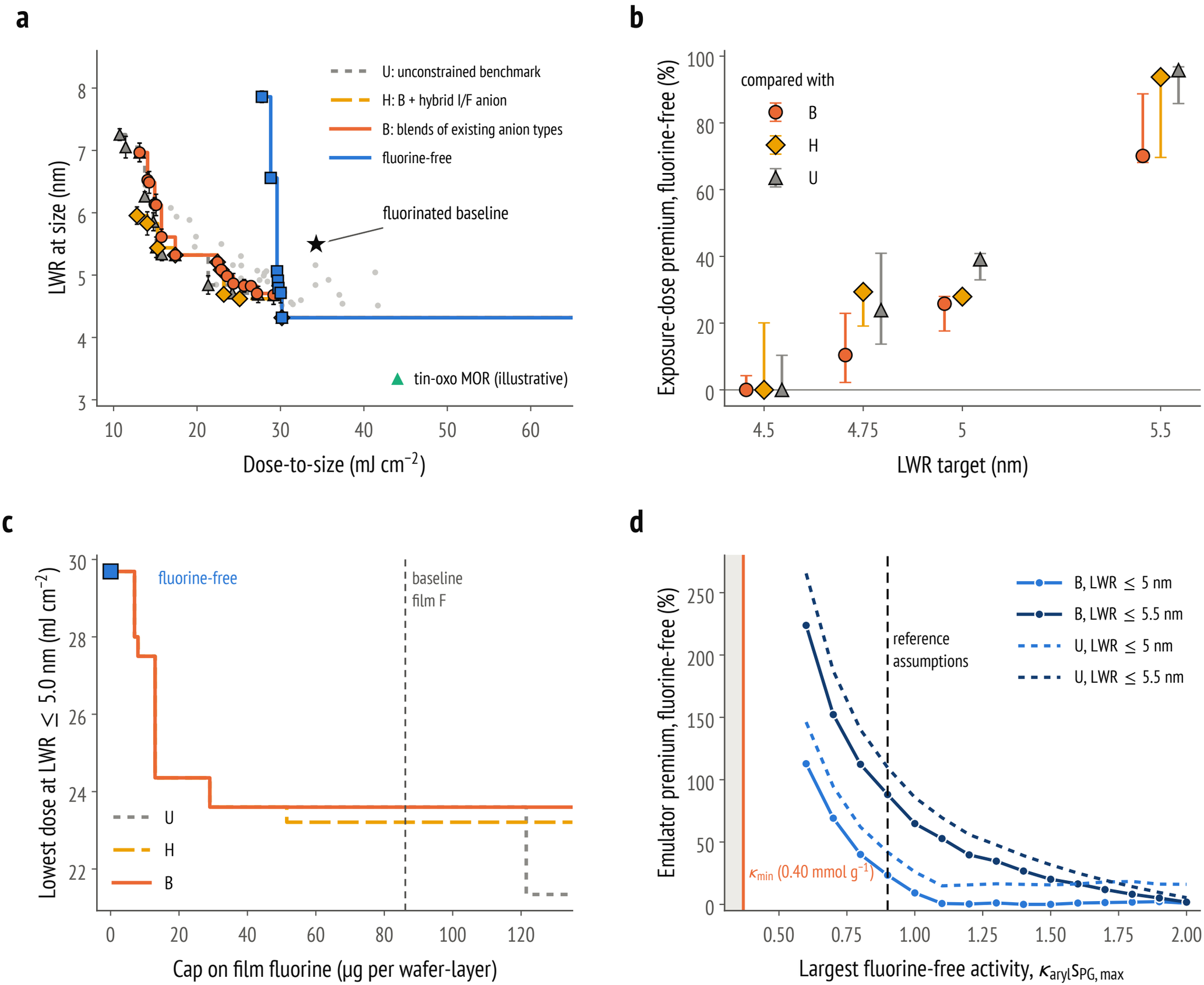
a
U: unconstrained benchmark
H: B + hybrid I/F anion
B: blends of existing anion types
fluorine-free
fluorinated baseline
tin-oxo MOR (illustrative)
LWR at size (nm)
Dose-to-size (mJ cm^{-2})
b
compared with
B
H
U
Exposure-dose premium, fluorine-free (%)
LWR target (nm)
c
fluorine-free
baseline
film F
U
H
B
Lowest dose at LWR ≤ 5.0 nm (mJ cm^{-2})
Cap on film fluorine (µg per wafer-layer)
d
B, LWR ≤ 5 nm
B, LWR ≤ 5.5 nm
U, LWR ≤ 5 nm
U, LWR ≤ 5.5 nm
reference
assumptions
κ_{min} (0.40 mmol g^{-1})
Emulator premium, fluorine-free (%)
Largest fluorine-free activity, $\kappa_{aryl}S_{PG,max}$